



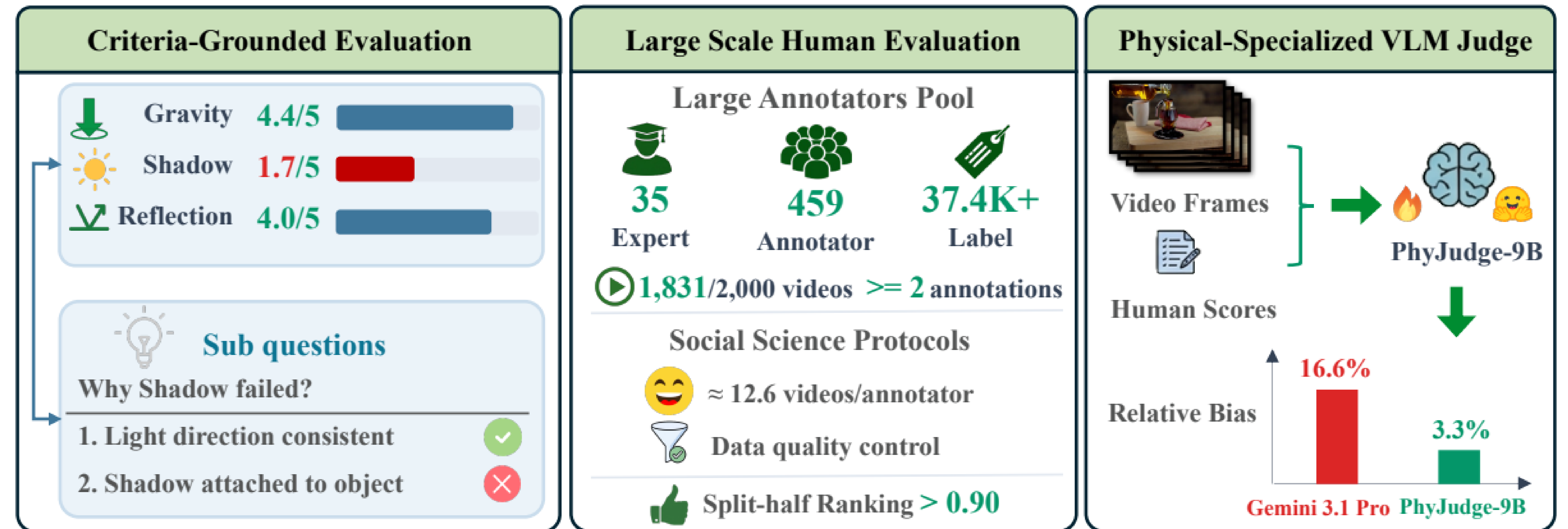
PhyGround: Benchmarking Physical Reasoning in Generative World Models

Juyi Lin



The Physics Problem

- **The Challenge:** Generative models create realistic videos but often violate basic physical laws, producing floating objects or disappearing fluids, undermining their real-world applicability.
- **Critical Evaluation Gaps:** Current benchmarks for evaluating physical reasoning in models are unreliable and lack diagnostic power due to three key flaws:
 - ***Coarse* Scores:** Domain-level ratings mask specific, law-level failures.
 - **Human Bias:** Small-scale annotation is vulnerable to fatigue and shortcuts.
 - ***Closed* Judges:** Non-reproducible VLM judges hinder auditable science.



Human Evaluation Results

Model	General Physics $\uparrow$			Domain Physics $\uparrow$			Overall $\uparrow$
	SA	PTV	Persist.	Solid-Body	Fluid	Optical	
Wan2.2-27B-A14B	<u>3.10</u>	3.37	3.50	3.23	3.18	<u>3.55</u>	3.28
Veo-3.1 [†]	3.26	3.29	3.42	3.12	3.65	3.69	3.28
OmniWeaving	2.97	3.17	3.34	2.98	<u>3.26</u>	3.22	3.10
Cosmos-14B	2.66	2.98	3.20	2.81	2.98	3.38	2.91
LTX-2.3-22B	2.58	2.74	2.72	2.62	3.07	2.98	2.69
Wan2.2-TI2V-5B	2.44	2.68	2.78	2.58	2.71	2.99	2.63
Cosmos-2B	2.33	2.56	2.77	2.53	2.77	3.35	2.58
LTX-2-19B	2.56	2.64	2.48	2.46	3.04	2.78	2.56

	Score	Spearman's ρ	Kendall τ -b
General	0.930 $\pm$ 0.045	0.828 $\pm$ 0.071	
Physics	0.907 $\pm$ 0.058	0.801 $\pm$ 0.092	
Overall	0.928 $\pm$ 0.051	0.830 $\pm$ 0.088	

Split-half ranking reliability

Benchmark	Pairwise agreement (%) $\uparrow$	N
Total score	73.0	58,984
WorldModelBench	70.0	—
VideoPhy-2	75–80	—

PhyGround: Our Solution

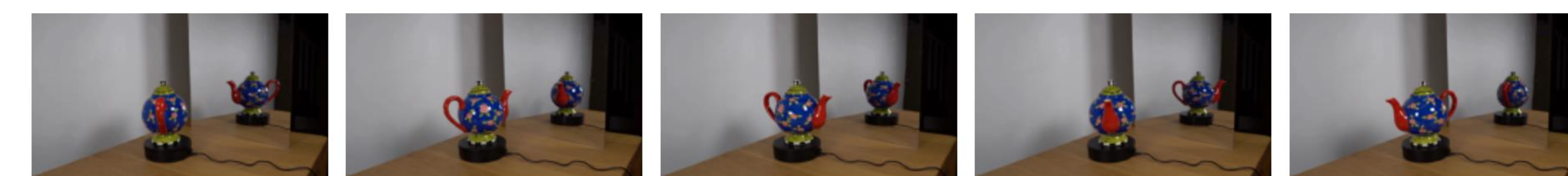
- Introducing **PhyGround**: A criteria-grounded benchmark for diagnosing physical reasoning in video generation.
- **Three Core Contributions:**
 1. **Fine-Grained Taxonomy:** A taxonomy of 13 physical laws enables precise, per-law diagnostic scores.
 2. **Large-Scale Human Study:** A quality-controlled study with 352 annotators provides a reliable evaluation foundation.
 3. **Open VLM Judge:** PhyJudge-9B is a reproducible, physics-specialized judge fine-tuned on our human annotations.

- **Diagnostic Value:** Models with similar overall scores show different strengths (e.g., one excels at solids, another at fluids/optics).
- **High Reliability:** Our large-scale study shows high stability and agreement (split-half Spearman's $\rho > 0.90$), creating a solid benchmarking foundation.

Benchmark & Judge Design

- **Expected-Outcome Prompts:** 250 prompts are augmented with *explicit* outcomes (e.g., '...and falls') to ensure clear, *unambiguous* evaluation.
- **Sub-Question Decomposition:** The VLM judge uses a sub-question checklist for each law to guide its reasoning and ensure objectivity.
- **PhyJudge-9B Training:** Fine-tuned on our human annotations using **LoRA**, creating a fully open, reproducible judge.

PhyJudge-9B Performance



Domain: Optical Model: Veo3.1
Prompt: A teapot on a rotating display base in front of a mirror; the reflection rotates in the opposite direction.

Judge	SA	PTV	Persist.	Reflection	Shadow	Overall
Human	4.0	4.3	4.3	4.7	4.3	4.4
PhyJudge	4	5	5	4	5	4.6
Gemini	3	5	2	1	1	2.2

Model	General (%) $\downarrow$	Physics (%) $\downarrow$	Overall (%) $\downarrow$	Signed (%) $\rightarrow 0$
Claude (Opus 4.7)	16.2	10.0	13.1	+14.5
Claude (Opus 4.7, CoT)	25.2	34.4	29.8	+34.3
Gemini-3.1-Pro	17.3	15.9	16.6	-6.3
Gemini-3.1-Pro (CoT)	17.2	17.4	17.3	-6.5
Qwen9B-base	28.2	35.0	31.6	+37.7
Qwen9B-base (CoT)	41.8	41.6	41.7	+47.8
Qwen27B-base	21.3	40.9	31.1	+35.7
Qwen27B-base (CoT)	37.3	46.9	42.1	+48.3
PhyJudge-9B	1.0	5.5	3.3	-1.1